Observationally Equivalent Algorithms: What Prices Reveal About Trader Populations, and What Supervisors Need

Pratik Niroula* Prashant Niroula † Sujan Khatiwada‡

13 November 2026

Abstract

Algorithmic traders read prices and, by trading, move them. One object, the feedback kernel, is a minimal sufficient statistic for such a market: populations with the same kernel produce the same prices, profits and equilibria, and prices identify nothing finer. Learning the composition of the population costs data of order $(\Delta\eta)^{-(4K-2)}$ for K groups whose memories lie within $\Delta\eta$, even near instability: two-group populations with half-lives of 5 to 9 days need more than 4,000 years of daily prices to be told apart. Market-wide actions need only the kernel, and a broad class of market-wide questions is answered at the usual statistical rate whatever the number of groups, so the two costs separate without bound as memories converge. When the population drifts, composition can be tracked only below a speed limit. Two remedies follow. Tagging all but one memory class removes the barrier, and the last tag is worth nothing. Without any tags, pooling D markets in which the same algorithms trade in different proportions lowers the exponent near generic configurations to $2(K - 1 + \lceil K/D \rceil)$, because an algorithm's memory is the same everywhere it trades. Core results are machine-checked in Lean 4.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, Lean proofs, manuscript drafting and internal critique. Human authors remain responsible for the content. AI checks do not

*Department of Computer Information Science, Minnesota State University, Mankato, MN 56001, USA. pratik.niroula@mnsu.edu.

†Faculty of Management, Tribhuvan University, Kirtipur, Nepal. prashantniroula20@gmail.com.

‡Department of Electrical, Computer and Robotics Engineering, Minnesota State University, Mankato, MN 56001, USA. sujan.khatiwada@mnsu.edu.

constitute independent review. The authors have verified all mathematical proofs, Lean 4 formalizations and empirical calculations, and take full responsibility for the content of the manuscript.

Supplement and companion. The online supplement states every result in full, with proofs and machine-check details. Results on cascades after a breach, insiders, and nonlinear or learning traders are in a companion note (Niroula et al., 2026c); references of the form C1.1 are to it.

Keywords: algorithmic trading, feedback trading, sufficient statistics, identification, supervision, disclosure. **JEL:** C32, D83, G14, G18.

Code and proofs. https://github.com/Pratikn03/Fianince.

1 Introduction

Trend-following funds, execution algorithms and AI trading agents read recent prices, decide what to trade and, by trading, move the prices they read. Regulators are now deciding how far to identify them. China's securities regulator issued rules on program trading in 2024, and since July 2025 its exchanges classify accounts as high-frequency by their order rates.[1] India requires every algorithmic order to carry an exchange-issued identifier.[2] Japan has registered high-speed traders since 2018, and its regulator studies their orders by trader and strategy.[3] The EU requires firms to identify their algorithms in order records under MiFID II. In the United States, the SEC withdrew its proposal on firms' use of predictive data analytics in June 2025. Behind these choices is a question this note answers: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

The answer. We make the intervention, not the trader, the unit of analysis, and ask what a supervisor must know for a given action to be safe. Prices reveal the market's *feedback kernel*, the response, m periods later, of total feedback flow to a unit move in the price signal. The kernel is all that prices identify, and it is all that anyone trading on public information needs. An action whose consequences depend on the population only through the kernel, such as a market-wide limit, can be set from prices at the usual statistical cost. The composition of the trader population matters only for actions aimed at one group, and learning it from prices is very expensive, even on the eve of a crash: at illustrative parameters, more than 4,000 years of daily prices when the traders' memories have half-lives between 5 and 9 days (Table 1). The two costs separate without bound: as the groups' memories approach each other, the cost of composition grows like $(\Delta\eta)^{-(4K-2)}$ while the cost of a broad class of market-wide questions does not move (Corollary 4.3). The expense does not come from the observer's tools: it holds for passive observation as well as for probing, and over any class of trading rules that contains the linear ones. This leaves a supervisor with a problem that the note states precisely at the end: keeping a market safe without knowing who trades in it.

Results.

(1) *Sufficiency* (Theorem 3.1). With a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, two populations with the same kernel give the same

[1] Shanghai, Shenzhen and Beijing stock exchanges, implementing rules of 3 April 2025 for the CSRC Provisions on Program Trading of May 2024.

[2] SEBI circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013, 4 February 2025.

[3] Financial Services Agency, FSA Analytical Notes, 21 November 2024.

prices, order flows, insider profits and dealer inventories, realization by realization. The two games therefore have the same equilibria.

(2) *Identification* (Theorem 3.2). Two populations give identical price responses to every trade sequence exactly when their *merged measures*, the trading weight at each memory rate, coincide. Groups with the same memory cannot be separated however long the record. For any population, linear or not, prices identify exactly its aggregate flow map, and the cost bounds below hold over any class that contains the linear rules.

(3) *Cost* (Theorem 4.1, Proposition 4.4). Separating K groups whose memories lie within $\Delta\eta$ takes of order $(\Delta\eta)^{-(4K-2)}$ experiments; Prony's method attains this order, and passive observation needs at least as much data. The barrier does not lift as the market approaches instability: prices then reveal how close the market is to breaking, not who is trading. Table 1 gives the data this takes at illustrative parameters.

(4) *Which questions are cheap* (Theorem 4.2). A linear question about the population is answered at the usual statistical rate, whatever the number of groups and the spacing of their memories, exactly when its kernel has square-summable power-series coefficients; its exact cost is a problem of polynomial approximation. Market-wide quantities are cheap; composition is not a linear question and is expensive.

(5) *Surveillance over time* (Theorem 6.1). When the population drifts, composition can be tracked only at the cube-root rate and not at all above a speed limit; traders who learn from prices impose a floor that no probing removes; and surveillance without tags pushes high-stakes algorithms together without ever detecting them.

(6) *Supervision* (Section 5). A uniform cut in feedback activity needs only the kernel. Halting the largest group needs composition, at the cost above. Class-tagged flow reports remove the cost, and tagging all but one memory class is enough: the cost is set by the untagged classes alone. Pooling markets in which the same algorithms trade in different proportions lowers the exponent without any tags. A group that can mimic another group's memory caps what any price-based targeting can achieve.

(7) *Limits* (Section 7). With position limits the sufficient object becomes a measure over memory and leverage. Calm prices reveal only its linear part, the kernel. Probes large enough to reach the limits identify the rest.

What is known. Exponential memory is the oldest model of expectations: adaptive expectations (Cagan, 1956; Nerlove, 1958), geometric distributed lags (Koyck, 1954), and Muth (1960)'s proof that exponentially weighted forecasts are optimal for a random walk observed

with noise. The idea that an aggregate depends on a distribution of individual coefficients and not on the individuals is also old. Granger (1980) showed it for sums of autoregressions, Forni and Lippi (1997) developed the general theory of aggregation and dynamics, Zaffaroni (2004) and Pesaran and Chudik (2014) the large-economy and large-panel versions, and recovering the distribution is the disaggregation problem (Celov et al., 2010). Results of the form "two structures give the same data, so data cannot choose" have a famous precedent in Sargent (1976), whose observational equivalence of natural and unnatural rate theories settled what reduced-form data could say about monetary rules. A large literature models markets with heterogeneous fundamentalist and chartist traders (Brock and Hommes, 1998; Hommes, 2006) and estimates their shares from prices (Boswijk et al., 2007; Franke and Westerhoff, 2012; Lux and Zwinkels, 2018); our results say which features of a population such estimates can recover from prices, and at what cost, when demand is linear in past prices. Auclert et al. (2024) make the corresponding sufficient-statistic argument for household heterogeneity in macroeconomics. That a decision needs only a reduction of the data is the subject of sufficiency and the comparison of experiments (Blackwell, 1953; Le Cam, 1964), and control theory distinguishes identifying a system from having data that are informative enough for a control objective (van Waarde et al., 2020). The rate $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for mixtures with near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020), and the uniqueness step behind the stress result is the uniqueness of networks with one hidden layer (Sussmann, 1992). We do not claim these.

What is new. The market setting, the policy question, and six results with no counterpart in the literatures above. (a) Sufficiency holds path by path inside a strategic market, for every insider and dealer rule, so it transfers to every equilibrium concept, including when impact and persistence come from an inventory equilibrium (Supplement Theorem S3.2). (b) The memory rates are transmission zeros of the closed-loop price response, not poles, which is why passive data are no cheaper than probing (Supplement Theorem S4.3). (c) The barrier does not lift near instability, where one expects the loop to amplify differences between populations (Proposition 4.4). (d) With partial disclosure the cost is set by the number of untagged memory classes, not traders, and the last tag is worth nothing (Theorem 5.3). (e) Without disclosure, the fact that an algorithm's memory is the same in every market it trades identifies it across markets (Theorem 5.4). (f) Every linear question about the population has an exact cost given by a problem of polynomial approximation, which sorts supervisory questions into cheap, boundary and expensive ones (Theorem 4.2). With position limits, calm prices carry exactly no information about how close traders are to their limits.

2 Model

Time is discrete. Let x_t be the pricing error, the gap between the price and the fundamental value. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$. There are K groups of feedback traders. Group i keeps an exponentially weighted average $y_{i,t}$ of past pricing errors and trades $q_i y_{i,t}$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \qquad y_{i,t} = (1 - \eta_i) y_{i,t-1} + \eta_i x_t, \tag{1}$$

where v_t is a probe, a trade by an observer, and ε_t is news. A desk that tracks a trend with a steady-state Kalman filter (Muth, 1960), or that learns by regularized online gradient descent, updates exactly this way (Supplement Lemma S1.1). Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the trading weight. The *merged measure* $M = \sum_i w_i \delta_{r_i}$ puts weight w_i at rate r_i, merging groups with equal rates, and the *feedback kernel* is its sequence of moments,

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots$$

Total feedback flow is the convolution of past prices with k. The *feedback budget* $S = a + \kappa \sum_i q_i = a + \kappa \sum_m k_m$ governs stability: with $q_i \geq 0$ the market is stable exactly when $S < 1$.

The observer. The cost results concern a supervisor or exchange that knows (a, κ) and the noise law. It can wait and record prices, probe from a reset state and record the response, probe the running market with random signs (which costs the same order of data, Supplement Theorem S16.1), or receive class-tagged flow reports. Reset probing is a benchmark for what data can reveal, not a proposed policy. Knowing (a, κ) only helps the observer, so the lower bounds below hold a fortiori when (a, κ) must be estimated.

When impact is mechanical. Model (1) lets feedback flow move the price. That describes markets in which prices are set by the mechanics of a limit order book or an automated market maker, or by dealers who price flow by its effect on their inventory (Supplement Theorem S3.1). It does not describe a market maker who sets prices as conditional expectations of value and knows the aggregate flow rule. Feedback flow is then predictable from past prices, is netted out and moves no price, so prices are those of a market without feedback traders and reveal nothing about them (Companion Proposition C2.1(iv)). Sufficiency (Theorem 3.1) holds for any dealer rule. The identification and cost results are stated for

mechanical impact; the barrier they describe is not an artefact of it, since with a fully rational market maker prices reveal less, not more.

3 The kernel is sufficient and is what prices identify

For sufficiency we embed the groups in a general market. A value v is drawn, noise traders submit z_t, an insider submits $X_t(v, \mathcal{H}_{t-1})$, and a dealer sets the price $P_t(\mathcal{H}_{t-1}, y_t)$ from the public history and the total order flow y_t. The groups update on a public signal $s_t = S_t(p_0, \ldots, p_t)$ and contribute flow $f_t = \sum_{m<t} k_m s_{t-1-m}$. The maps X, P and S are arbitrary causal rules, which covers Kyle, inventory-averse, learning and misspecified dealers.

Theorem 3.1 (The kernel is sufficient)**.** *If two group configurations have $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on dates $t \leq T$. The two games have the same strategy sets and payoffs, and hence the same equilibria under any concept defined from payoffs and the public history. The same holds for any finite population whose flows are causal linear functions of the public signal, with k the sum of the individual kernels. Group-level quantities, such as the weight of the largest group, are not invariant.*

The proof is an induction on dates (Supplement Theorem S2.1 and Proposition S2.2). It is short; its content is the object it isolates, a population-level quantity that settles every equilibrium concept at once. The induction uses only that aggregate feedback flow is a fixed causal function of the public signal. So for any population, linear or not, including traders who switch strategies after observing past profits, the aggregate flow map is sufficient (machine-checked in this generality; Supplement Theorem S6.1); the kernel is that map when flows are linear. For the same reason, observing aggregate order flow without class tags adds nothing about composition.

Theorem 3.2 (What prices identify)**.** *In model (1), two economies produce identical price responses to every probe sequence if and only if their merged measures coincide.*

This is Supplement Theorem S2.3(i). The same holds when κ and a are derived from competitive dealers who dislike inventory: the equilibrium price follows (1) path by path with effective impact and persistence, and with the memory rates unchanged (Supplement Theorems S3.1 and S3.2).

Together, Theorems 3.1 and 3.2 make the kernel a *minimal* sufficient statistic of the population for prices: it determines everything prices can show, and prices determine it. (The merged measure has finitely many atoms, so its moments, the kernel, determine it.)

Any population. For any population, linear or not, probe responses identify exactly the aggregate flow map, and a group-level quantity is identified if and only if it is a function of that map (Supplement Theorem S6.7, machine-checked for an arbitrary flow map). Theorem 3.2 is the linear case. The lower bounds of the next section hold over any class of rules that contains the linear ones, so richer rules cannot make composition cheaper to learn.

Proposition 3.3 (What an intervention needs)**.** *Let observations depend on the population only through its aggregate flow map, as prices and probe responses do.*

(i) If the loss from an action depends on the population only through the flow map, the set of safe actions is a function of what prices identify.

(ii) If two populations have the same flow map and an action is safe in one and unsafe in the other, every rule based on observations takes the action with the same probability in both.

(iii) If the records of the two populations are within total-variation distance δ, a rule that takes the action with probability p where it is safe takes it with probability at least $p - \delta$ where it is unsafe.

Parts (i) and (ii) restate Theorem 3.1, and part (iii) is the definition of total variation. The proposition is the decision form of sufficiency (Blackwell, 1953; Le Cam, 1964); we state it because it organizes what follows. A uniform cut in feedback is case (i) and a halt of the largest group is case (ii) (Proposition 5.1). Theorem 4.1 bounds δ in case (iii) for composition: with N records, $\delta \le C\sqrt{N}(\Delta\eta)^{2K-1}/\sigma$.

4 The price of composition

Theorem 3.2 says that groups with different memories can be told apart in principle. The question is how much data that takes when memories are close.

Theorem 4.1 (The cost of composition)**.** *Fix $K \ge 2$, a center rate r_c, a total weight W and a spacing $\Delta\eta$, with every candidate economy stable. Each reset experiment records the noisy impulse response with Gaussian noise of variance σ^2.*

(i) Lower bound. *There are two K-group economies with rates in $r_c + \Delta\eta[-1, 1]$ and positive weights whose merged measures have the same moments of order $0, \dots, 2K-2$ and whose impulse responses differ by order $(\Delta\eta)^{2K-1}$. Telling them apart with error probability below $1/4$ needs $N \ge c\,\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(ii) Upper bound. *On any class of separated rescaled rates with weights bounded away from zero, Prony's method applied to the first $2K$ estimated kernel values recovers the weights from $N = C\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(iii) Waiting is no cheaper. *If prices are instead observed passively, driven by Gaussian news, the information per observation between the pair in (i) is of order $(\Delta\eta)^{4K-2}$, so at least of order $(\Delta\eta)^{-(4K-2)}$ observations are needed.*

Parts (i)–(iii) are Supplement Theorems S4.1, S4.2 and S4.4. Part (iii) does not weaken as the market approaches instability (Proposition 4.4). For two groups, halving the spread of memories multiplies the data needed by 64; for three groups, by 1,024. The pair in (i) consists of the Gauss and Gauss–Radau quadrature rules, which match moments as far as K nodes allow. Part (iii) uses that the memory rates are zeros of the closed-loop response, so passive data face the same near-cancellation as probes. Gaussian news is the least favourable case: for any news law whose location divergence is quadratically bounded, the divergence is at most $\sigma^2 I_f^*$ times the Gaussian one, so the order in (iii) is unchanged (Supplement Proposition S16.2; $\sigma^2 I_f^* = 2$ for Laplace news).

Four extensions. *Without resets.* A supervisor that trades random signs into a market running from an unknown state, and correlates prices with its own trades, estimates the response with error of order $T^{-1/2}$ whatever the law of the news; Prony's method then recovers the weights from order $(\Delta\eta)^{-(4K-2)}$ periods, and no method needs fewer (Supplement Theorem S16.1). *Without separation.* If groups may nearly coincide or be tiny, individual weights cannot be recovered uniformly, but the merged memory distribution can, at the same order $N^{-1/(4K-2)}$ in Wasserstein distance (Supplement Theorem S19.1). *General memory.* If memory is any mixture of exponentials, finite or infinite, composition is harder still: the best attainable accuracy for the memory distribution improves only like $1/\log(N/\sigma^2)$, and this order is attained, so halving the error squares the data needed (Supplement Theorem S18.2; both bounds are machine-checked, with larger constants). *Money.* Probing a market is trading in it: to tell one group from two with memories $\pm\delta$ apart, the quantity traded must grow like δ^{-2} and the capital at risk like δ^{-4} (Supplement Theorem S10.1 and Section S10). Priced this way, the cost of learning composition from prices is a lower bound on the money value of class tags to a supervisor who needs composition (Supplement Corollary S16.3).

Which questions are cheap. Composition is one question among many that a supervisor can ask of the merged measure. Consider a linear question $F_g(M) = \int g\, dM$, with g continuous, observed through the first L kernel taps with noise of standard deviation σ in each of N experiments (for example, aggregate feedback flow while a probe holds the price

at a unit impulse), over populations with memories at most $\bar{r} < 1$ and total weight at most $\bar{W}$. Write $\mathcal{R}(h) = (\sup h)_+ + (-\inf h)_+$.

Theorem 4.2 (Which questions prices answer cheaply). *(i)* Exact cost. *The number of experiments needed to answer F_g to accuracy ϵ is determined, up to a factor of about* 1.2 *in accuracy at the* 95% *level, by the value of a problem of polynomial approximation, $\min_c\{\delta\|c\|_2 + \bar{W}\mathcal{R}(g - \sum_{t<L} c_t r^t)\}$ at δ of order $\sigma N^{-1/2}$, in which each coefficient is penalized because it multiplies a noisy tap.*

(ii) Cheap questions. *The cost is of order $\sigma^2\epsilon^{-2}$, whatever the number of groups and the spacing of their memories, exactly when g is a power series with square-summable coefficients on $[0, \bar{r}]$.*

(iii) Impossible questions. *Without a memory bound, a question with $|g(r)|\sqrt{1-r}$ unbounded as $r \uparrow 1$ cannot be answered to any accuracy from any number of experiments.*

This is Supplement Theorem S14.1 and Proposition S14.2, adapted from the first author's thesis; part (i) specializes the modulus theory of linear functionals (Donoho, 1994) to this market, and the polynomial-approximation form is what the market adds. The theorem sorts the questions of this paper. A single tap and the discounted feedback $\int(1-\beta r)^{-1}dM$, $\beta < 1$, are cheap. The total feedback weight $\Phi = \sum_m k_m$, which with a and κ fixes the feedback budget and so stability, sits on the boundary and costs $\sigma^2\epsilon^{-2}\log(1/\epsilon)$, with a logarithm that cannot be removed (Niroula et al., 2026b). The weight of a group is not a linear functional of M at all, and costs $(\Delta\eta)^{-(4K-2)}$ (Theorem 4.1). Market-wide questions are cheap; questions about who trades are not.

Corollary 4.3 (Identification and market-wide questions separate). *Fix $K \geq 2$, an accuracy ϵ and a cheap question F_g as in Theorem 4.2(ii). Over K-group populations whose memories lie within $\Delta\eta$, the number of experiments needed to learn the weights to within a fixed fraction of total weight, divided by the number needed to answer F_g to accuracy ϵ, is at least $c\,\epsilon^2(\Delta\eta)^{-(4K-2)}$. It tends to infinity as $\Delta\eta \to 0$.*

The corollary combines Theorem 4.1(i), whose two economies have weights that differ by a fixed fraction of the total, with Theorem 4.2(ii), whose cost does not depend on the spacing. The same holds, with a logarithm, for the total feedback weight. In a Monte Carlo with two groups, the error of the discounted feedback does not move with the spacing of memories (fitted slope -0.02), while the error of the weights grows like its inverse cube (slope -3.02).

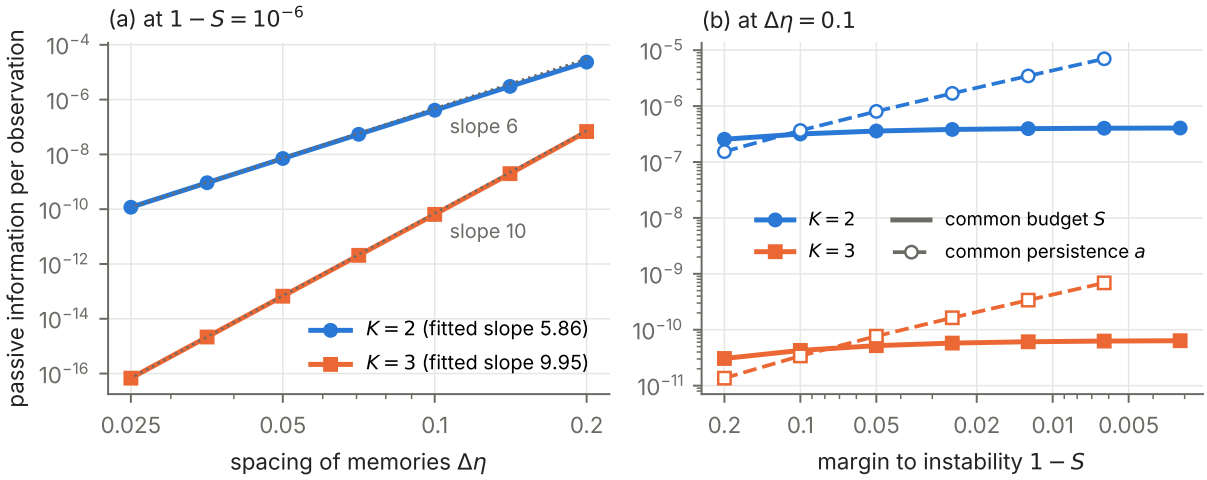


Figure 1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the K-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

Proposition 4.4 (The barrier does not lift near instability)**.** *Hold the pair of Theorem 4.1(i) at a common feedback budget $S = 1 - \epsilon$. The passive information per observation between them is at most $C(\Delta\eta)^{4K-2}$, with C independent of ϵ, however close the market is to instability.*

This is Supplement Proposition S4.5. One might expect a market near instability to reveal its traders, since the loop amplifies every difference. It reveals instead how close it is to instability: with a common persistence rather than a common budget, the information grows without bound as $\epsilon \to 0$, and all of the growth comes from the budget. The bound is per observation, and it extends to finite samples uniformly in ϵ (Supplement Proposition S12.5); in every case computed the exact information in n observations lies within 3% below n times the per-observation rate (Supplement Section S12.2). Figure 1 shows both statements.

How large is the barrier? Table 1 puts numbers on part (iii) for two groups. What matters is the spread of memories relative to their size. Populations whose memories all have half-lives between 5 and 9 days cannot be told apart with both error probabilities at 5% from less than 4,000 years of daily prices, however close the market is to instability.

Table 1: Years of daily price data below which no test tells the two populations of Theorem 4.1(i) apart with both error probabilities at 5%.

	memory rates	half-lives (days)	$1 - S = 0.1$	$1 - S = 10^{-6}$
$K = 2$	0.5 ± 0.1	0.8–1.4	20,000	16,000
	0.5 ± 0.05	0.9–1.2	1.2×10^6	9.0×10^5
	0.8 ± 0.1	1.9–6.6	100	69
	0.8 ± 0.05	2.4–4.3	6,000	3,900
	0.9 ± 0.05	4.3–13.5	120	77
	0.9 ± 0.025	5.2–8.9	7,000	4,300
$K = 3$	0.5 ± 0.1	0.8–1.4	1.5×10^8	9.9×10^7
	0.5 ± 0.05	0.9–1.2	1.5×10^{11}	9.6×10^{10}

Bound: $n \geq \ln 5/\mathrm{KL}$ observations (Bretagnolle–Huber), with KL the Gaussian information per observation, computed as in Figure 1, at a common feedback budget S, impact $\kappa = 0.5$ and feedback share $\kappa \sum_i q_i = 0.3$ (total weight $0.6(1 - r_c)$ at centre rate r_c). Memory rates are per day, 252 days a year; for traders whose memories are counted in five-minute periods, with prices recorded every five minutes around the clock, divide by 417. Two significant digits. Script: `research_extension/fragility_reveals/calibration_table.py`.

When half-lives differ by a factor of three or more, a century of daily data can suffice. With three groups the numbers grow to the order of 10^8 years. The numbers bound every test, for the passive record; probing faces the same order (part (i)). The exponents do not depend on Gaussian noise: with real Bitcoin price noise injected into synthetic groups, the fitted slopes are -6.01 and -9.81, against -6 and -10 (Supplement Section S22).

5 Who needs composition, and what disclosure buys

Proposition 5.1 (Market-wide versus targeted action)**.** *Let $q_i \geq 0$, let the market be stable with (a, κ) known, and let a supervisor need the feedback budget below a ceiling $\bar{s}$, with $a < \bar{s} < 1$.*

(i) Market-wide. *Scaling all feedback activity by θ meets the ceiling exactly when $a + \theta(S - a) \leq \bar{s}$. This depends on the population only through S, and so only through the kernel.*

(ii) Targeted. *Halting the largest group does not. For every $K \geq 2$ there are two populations whose merged measures have the same moments of order $0, \ldots, 2K - 2$, so that their budgets differ only by order $(\Delta\eta)^{2K-1}$, while their largest groups differ by a fixed amount, so that for a range of ceilings the halt meets the ceiling in one and not the other. Making this decision from N reset records with both errors at most $1/4$ needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$.*

Part (i) is Supplement Proposition S5.1(i). Part (ii) is Supplement Proposition S12.4; its pair comes from a family of quadrature rules whose weights are the values of one function

at their nodes (Supplement Lemma S12.3).

Theorem 5.2 (Value of class-tagged disclosure)**.** *Suppose that, in addition to prices, the observer receives each class's flow with Gaussian measurement noise. On the rich class of Theorem 4.1(ii), the number of experiments needed to recover the weights falls from order* $(\Delta\eta)^{-(4K-2)}$ *to a bound that does not depend on* $\Delta\eta$*. For the rates it falls to order* $(\Delta\eta)^{-2}$*.*

This is Supplement Theorem S5.3; Supplement Proposition S5.2 gives the regression that achieves it, one class at a time, with a variance that depends only on that class. Supplement Proposition S27.3 says when a supervisor should pay for such an audit trail: when the algorithms to be told apart are close enough that prices take too long, but not so close that even tagged data cannot help, and when the trail covers enough markets to recover its fixed cost.

Real regimes tag some traders and leave the rest anonymous. The next result prices that.

Theorem 5.3 (Partial disclosure)**.** *If the flows of* m *of the* K *groups are tagged, the cost of composition is that of the* $K' = K - m$ *untagged groups alone: of order* $(\Delta\eta)^{-(4K'-2)}$ *experiments if* $K' \geq 2$*, and a number that does not depend on* $\Delta\eta$ *if* $K' \leq 1$*.*

This is Supplement Theorem S5.4; the lower bound uses the K'-node Gauss and Gauss–Radau pair with the tagged groups held fixed, and the upper bound subtracts the tagged groups' estimated contribution and applies Prony's method to the rest. Each tag lowers the exponent by four. The tag that leaves a single group untagged is worth the most, a factor of order $(\Delta\eta)^{-6}$, and the last tag is worth nothing for the weights. So a supervisor never needs to tag everyone: tagging all but one memory class gives the full composition at a cost that does not depend on how close the memories are. What matters is how many distinct memory classes stay untagged, not how many traders. A regime that tags registered high-speed traders gets the full benefit only if everyone else behaves, in memory, like a single class.

A supervisor who sees several markets has a second route that needs no tags at all. An algorithm's memory, its lookback, is the same in every market it trades, while its participation varies from market to market.

Theorem 5.4 (Pooling markets)**.** *Let the same* K *algorithms trade in* D *markets, with common memory rates and market-specific weights whose matrix has rank* $D \leq K$*. Pooling the price records of the* D *markets, the data needed to learn composition are of order* $(\Delta\eta)^{-2(K-1+\lceil K/D\rceil)}$ *near generic configurations:* $(\Delta\eta)^{-(4K-2)}$ *with one market and* $(\Delta\eta)^{-2K}$ *with* K *or more. Over a compact class the exponent lies between* $2(K-1+\lceil K/D\rceil)$ *and* $4K-2$*, so pooling never needs more data than one market; whether it equals the local exponent is a global question that is open in general.*

This is Supplement Theorem S5.5. The lower bound on the joint records of all markets is machine-checked for any matched pair, and so is the upper bound given the rank condition, which holds at almost every configuration for any number of algorithms and markets (machine-checked at an explicit configuration); Monte Carlo runs of the estimator reproduce the predicted rates. Cross-market surveillance is a partial substitute for tagging, and the converse constrains hiding: to stay as hidden as in a single market, the algorithms must trade in the same proportions everywhere. For two algorithms with half-lives between 5 and 9 days, about 7,000 years of daily prices in one market become at least about 90 in two markets in which their shares differ by 50 percentage points (hardest pair found numerically; Supplement Section S5.3).

Two further results say what the supervisor cannot use instead. Markets with the same eigenvalues, the same one-step and long-run price variance can have arbitrarily different worst-case short-run gains, so no monitor built on eigenvalues and second moments of prices is uniformly safe (Supplement Theorem S26.1). And what prices hide is economically specific: for desks that track a trend with Kalman filters, prices depend on the desks only through their signal-to-noise ratios and intensities, so how good each desk's information is, the thing a supervisor would most like to know, is what costs $(\Delta\eta)^{-(4K-2)}$ to learn (Supplement Propositions S23.1 and S23.2).

The results so far take the groups' rules as given. A group that expects to be targeted has a reason to change them.

Proposition 5.5 (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for any record length, and then targets a given group or not. Being targeted costs the group $P > 0$. Suppose the group can adopt, at private cost C_0, a rule that gives the population the kernel of a population that the supervisor targets with probability α. Then in any best response the group is targeted with probability at most $\alpha + C_0/P$, whatever the probes, record length and decision rule.*

The proof is two lines from Theorem 3.1 (Supplement Proposition S5.6). More data make the group's original rule easier to detect and so make hiding more attractive. With several groups, the bound applies to each group's best response given the others' rules, so it holds group by group in any equilibrium; what determines each group's α is not modelled. A class tag is attached to the account, so changing the trading rule does not remove it.

6 Surveillance over time

Populations drift: algorithms enter, exit and are retuned. Suppose the population moves by at most v per period in its rescaled memories and weights, and the supervisor runs n reset experiments per period.

Theorem 6.1 (A speed limit on surveillance)**.** *With $\Gamma = \sigma^2 v(\Delta\eta)^{-(4K-2)}/n$, the best attainable error in tracking the weights is of order $\min(1, \Gamma^{1/3})$: no estimator does better along some admissible path, and Prony's method on a moving window of the right length attains it. If v exceeds a speed limit of order $n(\Delta\eta)^{4K-2}/\sigma^2$, no record of any length tracks the weights at all. A linear question with square-summable coefficients whose answer moves by at most v_F per period is tracked with error of order $(\sigma^2 v_F/n)^{1/3}$, whatever the number of groups and the spacing of their memories.*

This is Supplement Theorem S15.1. Composition is untrackable unless it is stable over the time it takes to identify it, which at illustrative parameters is thousands of years of daily prices; market-wide quantities are tracked at the cube-root rate. In a simulation with two groups moving along the hard family, the weight error grows with speed and spacing at fitted slopes 0.33 and -2.01, against $1/3$ and -2.

Two strategic responses sharpen the point. If traders adjust their rules in response to what they earn, a probe of size A buys information at rate A^2 and moves its target at rate A^2; for any class of learning rules whose response to probes is bounded by γ but otherwise unknown, every probing policy leaves a tracking error of at least order $\min(1, (\sigma^2\gamma(\Delta\eta)^{-(4K-2)})^{1/3})$ (Supplement Proposition S15.2). And a group that expects to be targeted once identified can move its memory toward its neighbour's. In a two-group game in which that move costs $k(D_0 - D)^2$ and detection costs L, more data without tags push the group's memory toward its neighbour's, and its detection probability never exceeds the false-alarm rate plus kD_0^2/L, however much data the supervisor has; tags remove the incentive to crowd (Supplement Proposition S15.3). Where algorithm identification is introduced, the dispersion of memories across algorithm classes should therefore rise, a prediction that only algorithm-level data can test.

7 Position limits: calm hides, stress reveals

Real feedback traders have position limits. Let trader i have memory rate r_i, capacity $c_i > 0$ and leverage ℓ_i, and trade $c_i\psi(\ell_i m_{i,t})$, where $m_{i,t}$ is its exponentially weighted signal and ψ is a shape common to all traders. The leading case is the hard limit $\psi(u) = \max(-1, \min(1, u))$:

inside its band a trader is linear with weight $c_i \ell_i$; at the edge its position is capped. Write $\nu = \sum_i c_i \delta_{(r_i, \ell_i)}$ for the capacity measure on memory and leverage.

Theorem 7.1 (Two layers). *(i)* Sufficiency. *For any shape, populations with the same ν give the same prices, profits and equilibria. The kernel is one linear functional of ν, $k_m = \int r^m \ell \, d\nu$, and does not determine it.*

(ii) Calm hides. *Under hard limits, two populations with the same kernel give identical price and order-flow paths until some trader first reaches its limit. Any record of the market before date T distinguishes them with total-variation distance at most the probability that a limit is reached before T. With Gaussian news and no probe this is at most $T(n+n')e^{-1/(2\vartheta^2)}$ for the $n+n'$ traders of the two populations, where ϑ is the largest standard deviation of a trader's position relative to its limit.*

(iii) Stress reveals. *If every trader has $r_i \geq 0$ and $\ell_i \neq 0$ and responses are observed exactly, probes of all sizes identify the signed capacity measure $\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i) \, \delta_{(r_i, |\ell_i|)}$, and nothing finer, for the hard limit and for* tanh*. For* tanh*, probes in any neighbourhood of zero suffice. For a cubic shape the result fails.*

Parts (i)–(iii) are Supplement Theorems S6.1, S6.2 and S6.4; for the hard limit and for tanh with probes of all sizes, part (iii) is machine-checked for any number of traders; the small-probe statement is proved on paper. The theorem assumes exactly observed responses; with observation noise, a minimum-distance estimator recovers the signed measure consistently (Supplement Proposition S12.6), and at the parametric rate for smooth shapes and, with probes that bracket the kinks, for the hard limit (Companion Propositions C3.3 and C3.4); information about the order-n leverage moment grows like the probe size to the power $2(2n+1)$ (Companion Proposition C3.2). Its last step, for tanh, is the uniqueness of one-hidden-layer networks up to sign flips (Sussmann, 1992), which is why only the signed measure is identified. What the market adds is the reduction: probing recovers the flow map from prices, and two price dates give different memory–leverage pairs different scales.

The economic content is in part (ii). How close traders are to their limits, which governs behaviour in a crash, leaves no trace in calm prices. It can be learned only by pushing traders toward their limits or from disclosure. It also decides whether a breach cascades: with mechanical impact, liquidation rounds settle smoothly when the liquidation triggered per unit of price move, $\kappa\ell$, is below one everywhere, and jump discontinuously where it exceeds one (Companion Theorem C1.1). Calm prices therefore reveal neither the cascade threshold nor which side of it the market is on. When dealers anticipate the liquidation but capital arrives slowly, the price falls at once, overshoots, and the threshold is lower (Companion Proposition C1.3).

8 Scope and conclusion

Evidence. The contribution is theoretical, and the tagged flows and position limits it concerns are rarely observed. Supplement Section S7 reports two registered tests on public data, on Taiwan's tagged exchange flows and on Binance perpetual futures, with their readouts as registered.

The sufficiency results hold whenever aggregate feedback flow is a causal function of public signals; the kernel is the sufficient object when that function is linear, and ν when it saturates as in Section 7. The explicit identification and cost results need linear rules with exponential memory. For general rules, prices identify the aggregate flow map and the lower bounds still hold (Supplement Theorem S6.7), and for rules within a bounded distance of linear, such as dead bands, large probes identify composition (Companion Proposition C3.1); for saturating families, probes that push traders toward their limits identify the signed capacity measure at the parametric rate, and composition costs the linear amount when groups share a leverage but an amount independent of their memory separation when leverages differ (Companion Proposition C3.5); for some nonlinear families composition is not identified at all. For linear rules with general completely monotone memory the minimax accuracy is of order $1/\log(N/\sigma^2)$ (Supplement Theorem S18.2). Simulations of saturating, adaptive and neural trading rules show where the linear theory holds and where it fails (Companion Section C4). The passive bounds are stated for Gaussian news, the least favourable case (Supplement Proposition S16.2); news with bounded support is not covered. Traders who learn from public data, their own trades and private randomness are covered for what prices reveal: prices identify their aggregate flow kernel and nothing finer, and linear learners whose parameters drift are tracked below the speed limit of Section 6 (Companion Proposition C3.6). A group that chooses its rule against an insider becomes a momentum trader that rides on her information, and its choice is still summarized by the kernel (Companion Proposition C2.2).

The problem this leaves. Together the results pose a problem. A supervisor who must keep a market stable cannot learn who trades in it from prices, at any reasonable cost, in calm conditions or near a crash; the traders can hide if it tries; and the information that would identify them arrives, if at all, only when limits are already being hit. Whether a market can be kept safe without that knowledge, using only what prices do reveal, is the question taken up in the companion paper (Niroula et al., 2026b).

What this says about the rules. The results sort the regulatory choices above. A market-wide measure, such as a uniform limit on feedback activity or a pause triggered by the feedback budget, needs no identification and can be set from prices. A measure

aimed at particular algorithms cannot be set from prices at any reasonable cost, and the targeted algorithms can make it harder still by imitating others. It needs flow tagged by class or account, which is what the Chinese, Indian, Japanese and European regimes collect. Whether that data make the targeted measures worth their cost is an empirical question that only those tags can answer.

Within these limits the message is simple. Prices tell a supervisor the kernel and nothing more, and the kernel is enough for any action that treats the market as a whole. Actions aimed at one group need composition, which prices supply only at a steep and known price and which a group can hide at a bounded cost. Disclosure tagged by trader class removes that price. The same split recurs with position limits: calm markets reveal the kernel, and only stress or disclosure reveals how close traders are to their limits.

Declarations

Author contributions. Pratik Niroula: conceptualization, methodology, formal analysis, software, investigation, data curation, writing (original draft). Prashant Niroula: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). Sujan Khatiwada: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). All authors approved the final manuscript.

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Data availability. All data are public. The Taiwan test uses the Taiwan Stock Exchange's daily reports of trading by the three major institutional investors (T86), closing quotes (MI_INDEX), ex-rights and ex-dividend adjustments (TWT49U) and market totals (FMTQIK), from https://www.twse.com.tw. The Binance test uses daily files of five-minute candles and trading metrics (including open interest) for USD-margined perpetual futures from the Binance public data archive (https://data.binance.vision). The registered specifications, amendments, executors, notebooks, result files and test-unlock counters are in the public repository.

Code availability. All code, Lean 4 proofs and result files are public at https://github.com/Pratikn03/Fianince.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Declaration of generative AI and AI-assisted technologies. During the preparation of this work the authors used Claude, ChatGPT/Codex and Gemini for research development, code, numerical checks, Lean proofs, drafting and internal critique, as stated on the title page. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Abramowitz, M., and I. A. Stegun (1964). *Handbook of Mathematical Functions*. National Bureau of Standards, Washington, DC.

Auclert, A., M. Rognlie, and L. Straub (2024). The intertemporal Keynesian cross. *Journal of Political Economy* 132, 4068–4121.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barroso, P., R. M. Edelen, and P. Karehnke (2022). Crowding and tail risk in momentum returns. *Journal of Financial and Quantitative Analysis* 57, 1313–1342.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Cagan, P. (1956). The monetary dynamics of hyperinflation. In M. Friedman (Ed.), *Studies in the Quantity Theory of Money*, 25–117. University of Chicago Press.

Celov, D., R. Leipus, and A. Philippe (2010). Asymptotic normality of the mixture density estimator in a disaggregation scheme. *Journal of Nonparametric Statistics* 22, 425–442.

Donoho, D. L. (1994). Statistical estimation and optimal recovery. *Annals of Statistics* 22, 238–270.

Dreesen, P., D. T. Westwick, J. Schoukens, and M. Ishteva (2017). Modeling parallel Wiener–Hammerstein systems using tensor decomposition of Volterra kernels. In *Latent Variable Analysis and Signal Separation (LVA/ICA 2017)*, Lecture Notes in Computer Science 10169, 16–25. Springer.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Forni, M., and M. Lippi (1997). *Aggregation and the Microfoundations of Dynamic Macroeconomics*. Oxford University Press.

Granger, C. W. J. (1980). Long memory relationships and the aggregation of dynamic models. *Journal of Econometrics* 14, 227–238.

Hale, N., and A. Townsend (2013). Fast and accurate computation of Gauss–Legendre and Gauss–Jacobi quadrature nodes and weights. *SIAM Journal on Scientific Computing* 35, A652–A674.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Koyck, L. M. (1954). *Distributed Lags and Investment Analysis*. North-Holland.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Li, W., Z. Zhu, W. Gao, and W. Liao (2022). Stability and super-resolution of MUSIC and ESPRIT for multi-snapshot spectral estimation. *IEEE Transactions on Signal Processing* 70, doi:10.1109/TSP.2022.3204454.

Muth, J. F. (1960). Optimal properties of exponentially weighted forecasts. *Journal of the American Statistical Association* 55, 299–306.

Nerlove, M. (1958). Adaptive expectations and cobweb phenomena. *Quarterly Journal of Economics* 72, 227–240.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Pesaran, M. H., and A. Chudik (2014). Aggregation in large dynamic panels. *Journal of Econometrics* 178, 273–285.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Sargent, T. J. (1976). The observational equivalence of natural and unnatural rate theories of macroeconomics. *Journal of Political Economy* 84, 631–640.

Sussmann, H. J. (1992). Uniqueness of the weights for minimal feedforward nets with a given input-output map. *Neural Networks* 5, 589–593.

Szegő, G. (1975). *Orthogonal Polynomials*, 4th ed. American Mathematical Society Colloquium Publications 23, Providence, RI.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.

Zaffaroni, P. (2004). Contemporaneous aggregation of linear dynamic models in large economies. *Journal of Econometrics* 120, 75–102.

Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics* 24(2), 265–272.

Le Cam, L. (1964). Sufficiency and approximate sufficiency. *Annals of Mathematical Statistics* 35(4), 1419–1455.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65(11), 4753–4768.

Boswijk, H. P., C. H. Hommes, and S. Manzan (2007). Behavioral heterogeneity in stock prices. *Journal of Economic Dynamics and Control* 31, 1938–1970.

Franke, R., and F. Westerhoff (2012). Structural stochastic volatility in asset pricing dynamics: Estimation and model contest. *Journal of Economic Dynamics and Control* 36, 1193–1211.

Hommes, C. H. (2006). Heterogeneous agent models in economics and finance. In L. Tesfatsion and K. L. Judd (eds.), *Handbook of Computational Economics*, Vol. 2, 1109–1186. Elsevier.

Lux, T., and R. C. J. Zwinkels (2018). Empirical validation of agent-based models. In C. Hommes and B. LeBaron (eds.), *Handbook of Computational Economics*, Vol. 4, 437–488. Elsevier.

Niroula, P., P. Niroula, and S. Khatiwada (2026a). What prices cannot tell a market supervisor: The cost of identifying trader groups and the value of class disclosure. Working paper, 5 November 2026, https://github.com/Pratikn03/Fianince.

Niroula, P., P. Niroula, and S. Khatiwada (2026c). After a breach, and beyond linear rules: Companion results to "Observationally equivalent algorithms". Working paper, 13 November 2026, https://github.com/Pratikn03/Fianince.